

Exam Seat No :

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B.E 8th SEMESTER EXAMINATION (July 2025)
SUBJECT: Neural Network and Deep Learning (Code: CT803B-N)

Date:16/07/2025

Time: 10:30 AM to 01:30 PM

Marks: 70

Instruction:

1. Answer each section in separate Answer Sheet.
2. All questions are compulsory.
3. Indicate **clearly**, the option you attempted along with its respective question number.
4. Use the last page of main supplementary for rough work.

Section:1

- Q.1** (a) What is Machine Learning? Explain its different types [05]
(b) What is need of Artificial Neural Network in today's era? [05]
(c) Explain applications of Neural Network. [05]

OR

- (c) What is the limitation of Back Propagation learning? Explain in details. [05]
Q.2 (a) What do you mean by CNN? Write down the advantages of CNN. [05]
(b) Draw and explain the LeNet architecture. [05]

OR

- Q.2** (a) Draw and explain the architecture of convolutional network. [05]
(b) Define Overfitting Problem with respect to AlexNet Architecture. [05]
Q.3 (a) What is Recurrent Neural Networks. Explain in details. [05]
(b) Which architecture network that Challenged CNNs? Given a briefly reason of it. [05]

OR

- Q.3** (a) Differentiate between Gradient Descent and Perceptron training rule. [05]
(b) Describe the role of convolutional layers, pooling layers, and activation functions in CNNs. Why are they essential? [05]

Section:2

- Q.4 (a) What are the limitation of McCulloch–Pitts unit neuron? [05]
(b) Describe the Convergence theorem for Perceptron Learning Algorithm. [05]
(c) What is Data Normalization, and Why Do We Need It? [05]

OR

- (c) List out the pros and cons of a typical RNN architecture. [05]
Q.5 (a) What is the difference between Machine Learning and Deep Learning? [05]
(b) What are autoencoders? [05]

OR

- Q.5 (a) How can hyperparameters be trained in neural networks? [05]
(b) What are Boltzmann Machines? And What is a Restricted Boltzmann Machine? [05]
Q.6 (a) Explain Gibbs Sampling in details. [05]
(b) Explain Generative Adversarial Network. [05]

OR

- Q.6 (a) What is role of Gradient descent in RBM? [05]
(b) Explain Role of Deep learning to the RBM. [05]